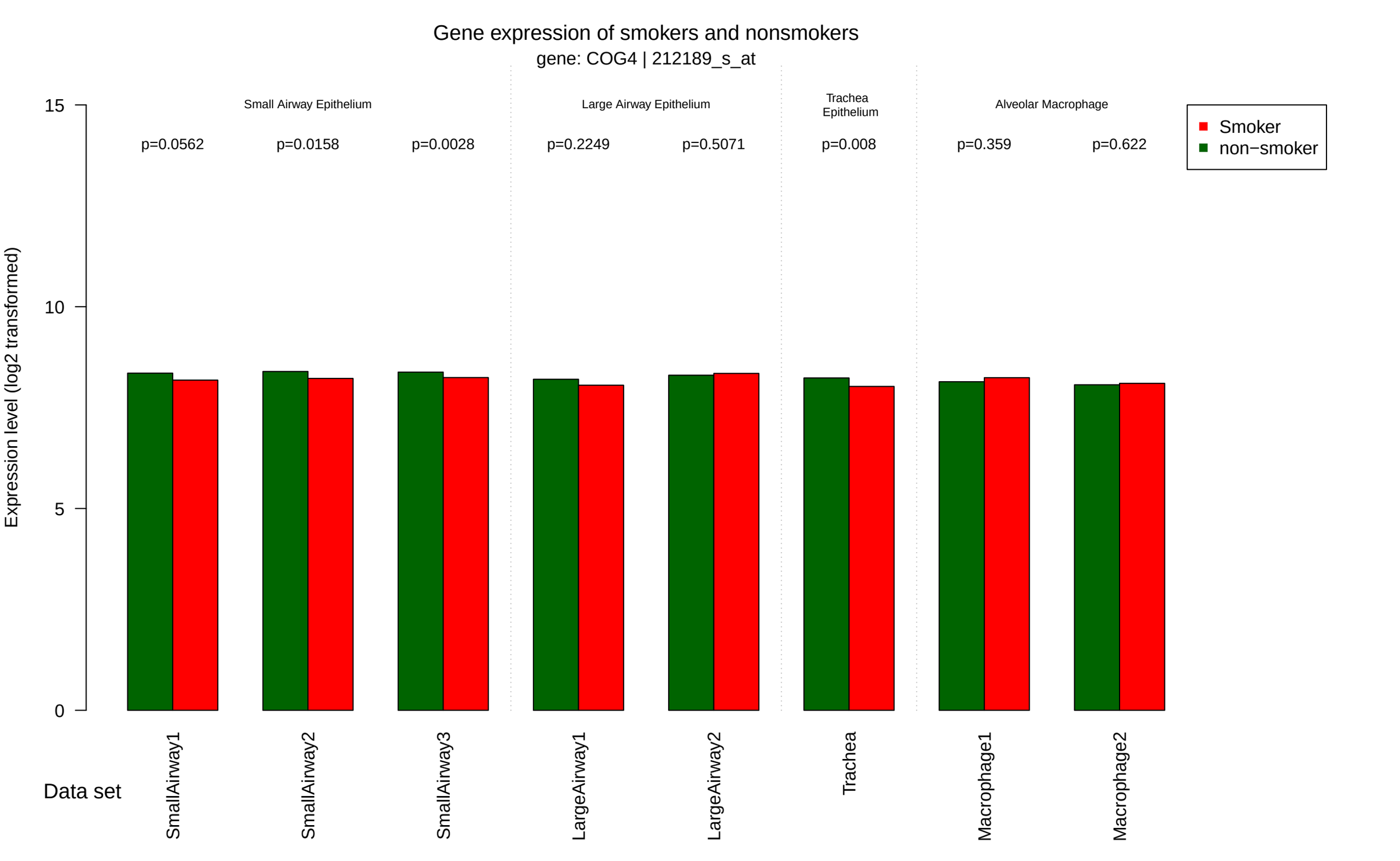
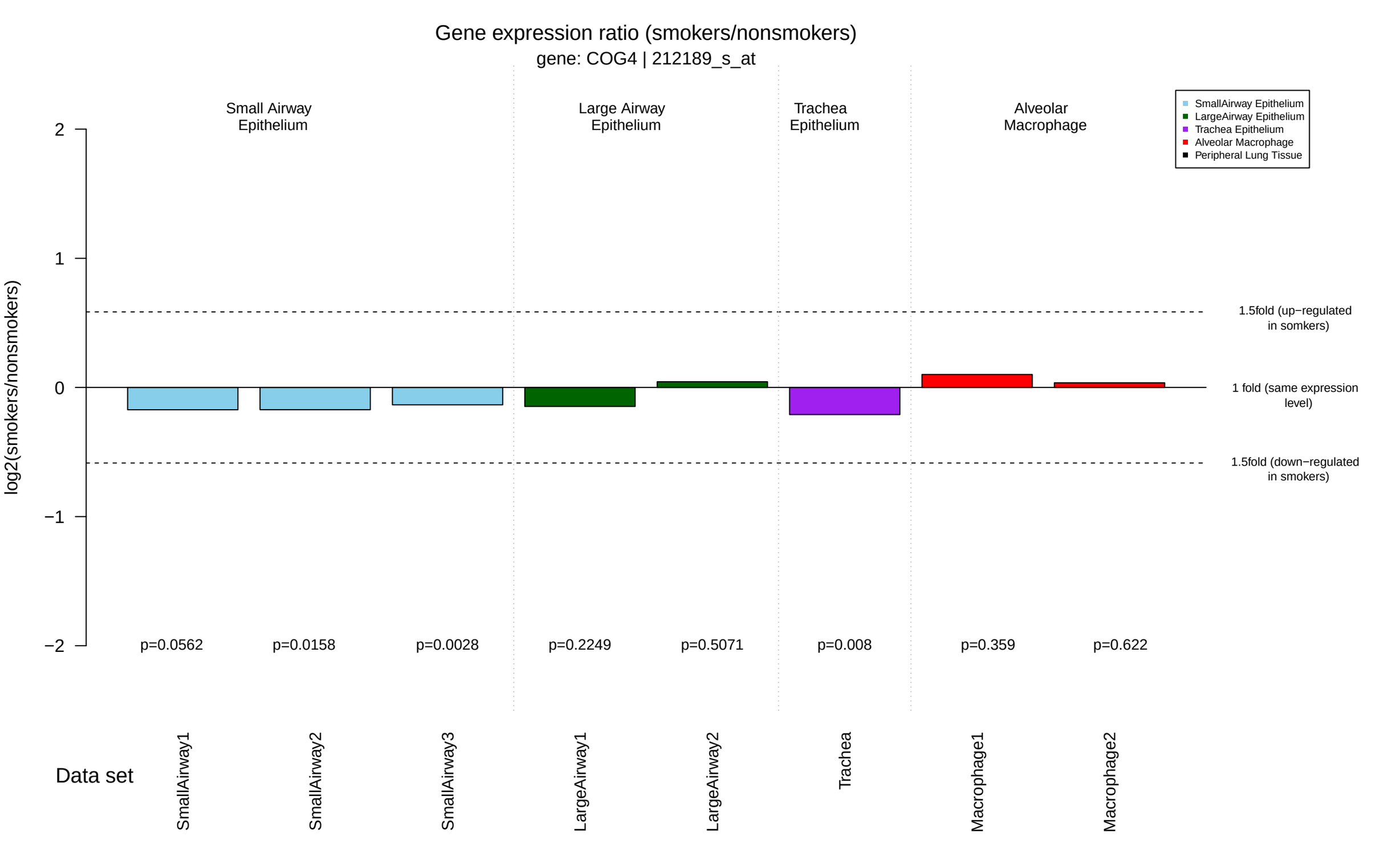
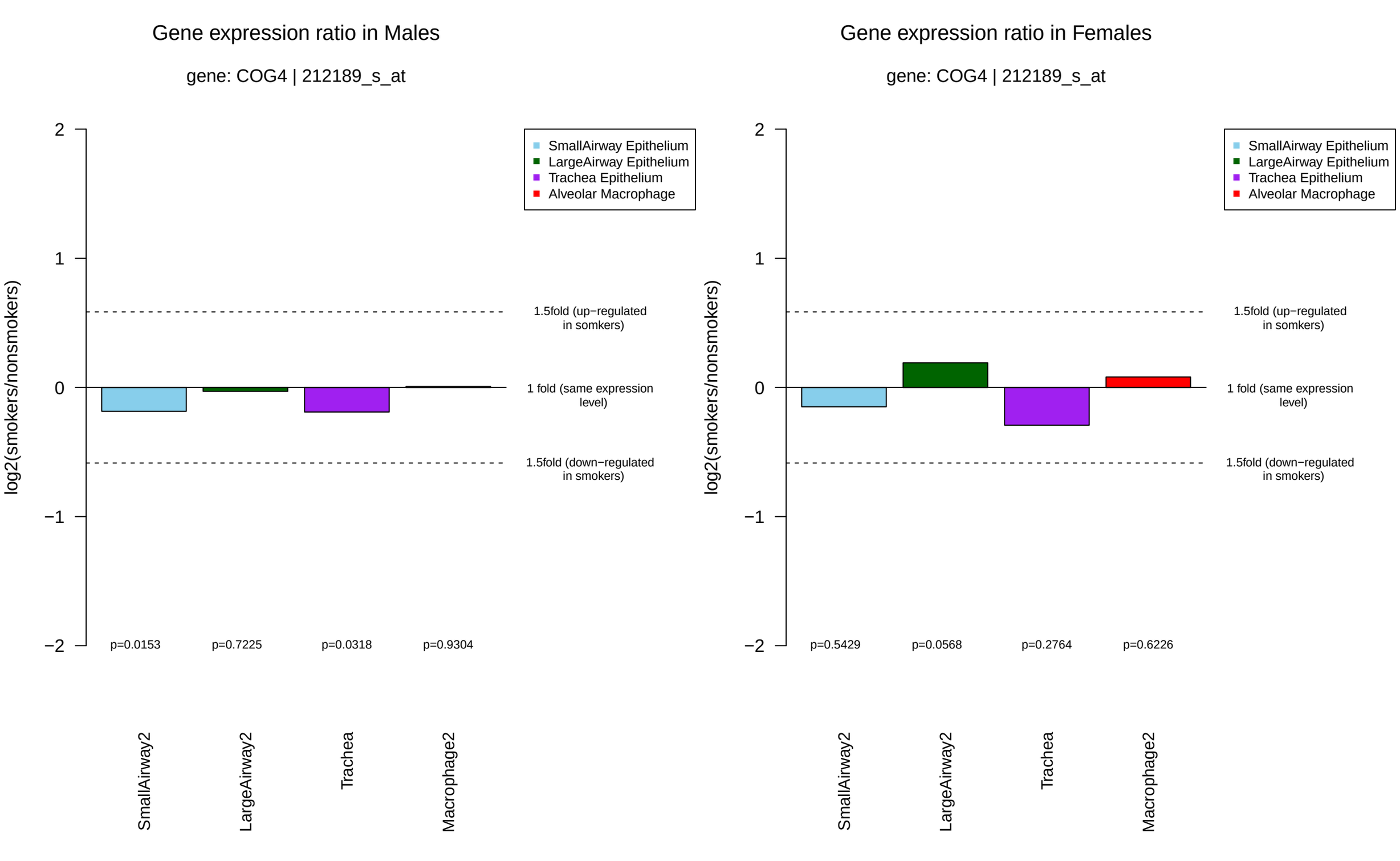
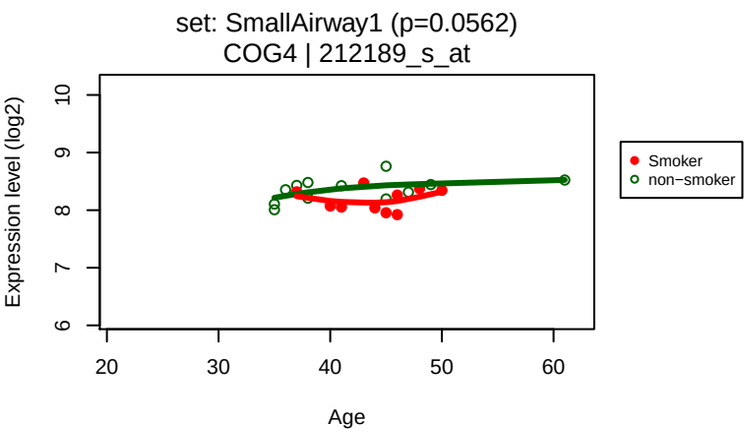
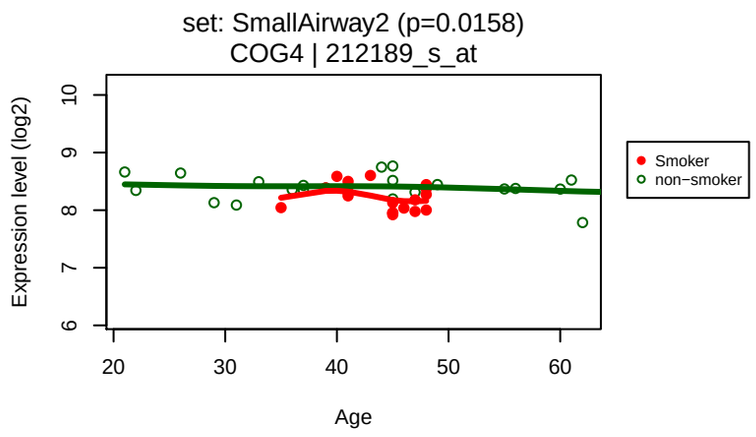
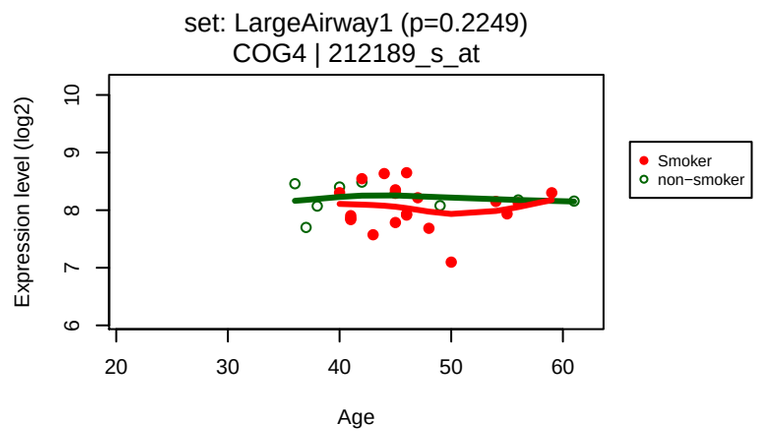

Gene expression of smokers and nonsmokers
gene: COG4 | 212189_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0562
p=0.0158
p=0.0028
p=0.2249
p=0.5071
p=0.008
p=0.359
p=0.622
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

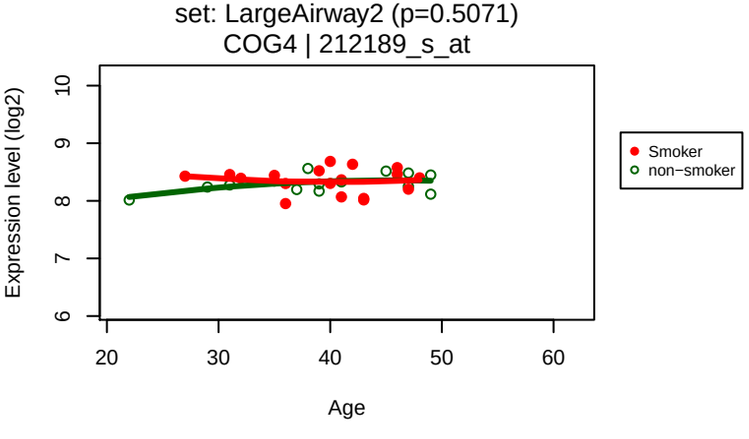

Gene expression ratio (smokers/nonsmokers)
gene: COG4 | 212189_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0562
p=0.0158
p=0.0028
p=0.2249
p=0.5071
p=0.008
p=0.359
p=0.622
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

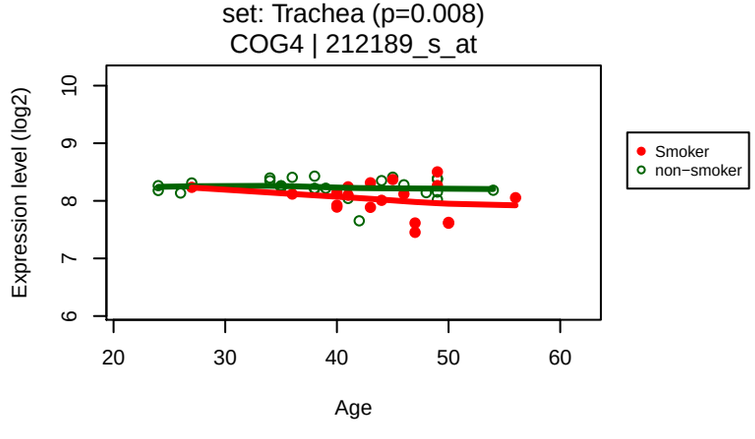

Gene expression ratio in Males
gene: COG4 | 212189_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0153
p=0.7225
p=0.0318
p=0.9304
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COG4 | 212189_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5429
p=0.0568
p=0.2764
p=0.6226
SmallAirway2
LargeAirway2
Trachea
Macrophage2

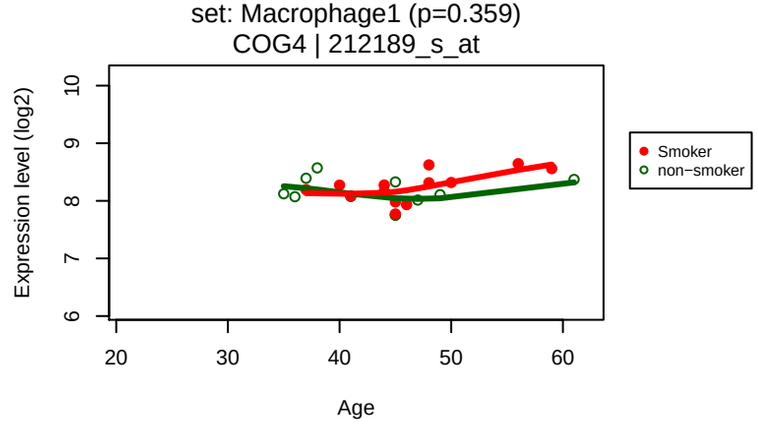

set: SmallAirway1 (p=0.0562)
COG4 | 212189_s_at
Expression level (log2)
Age
Smoker
non−smoker

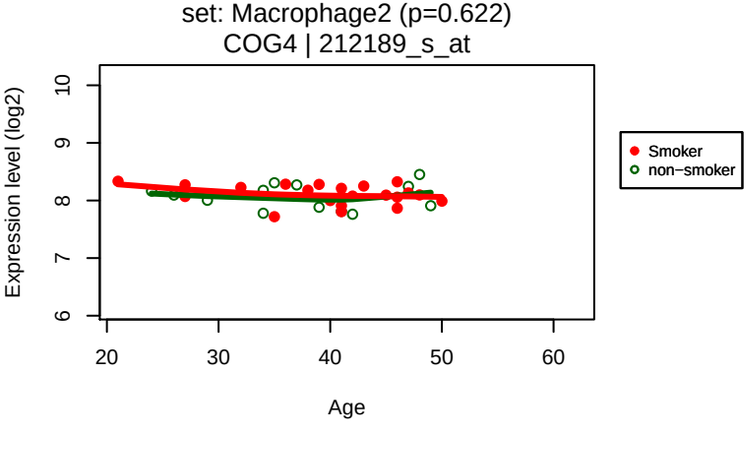

set: SmallAirway2 (p=0.0158)
COG4 | 212189_s_at
Expression level (log2)
Age
Smoker
non−smoker

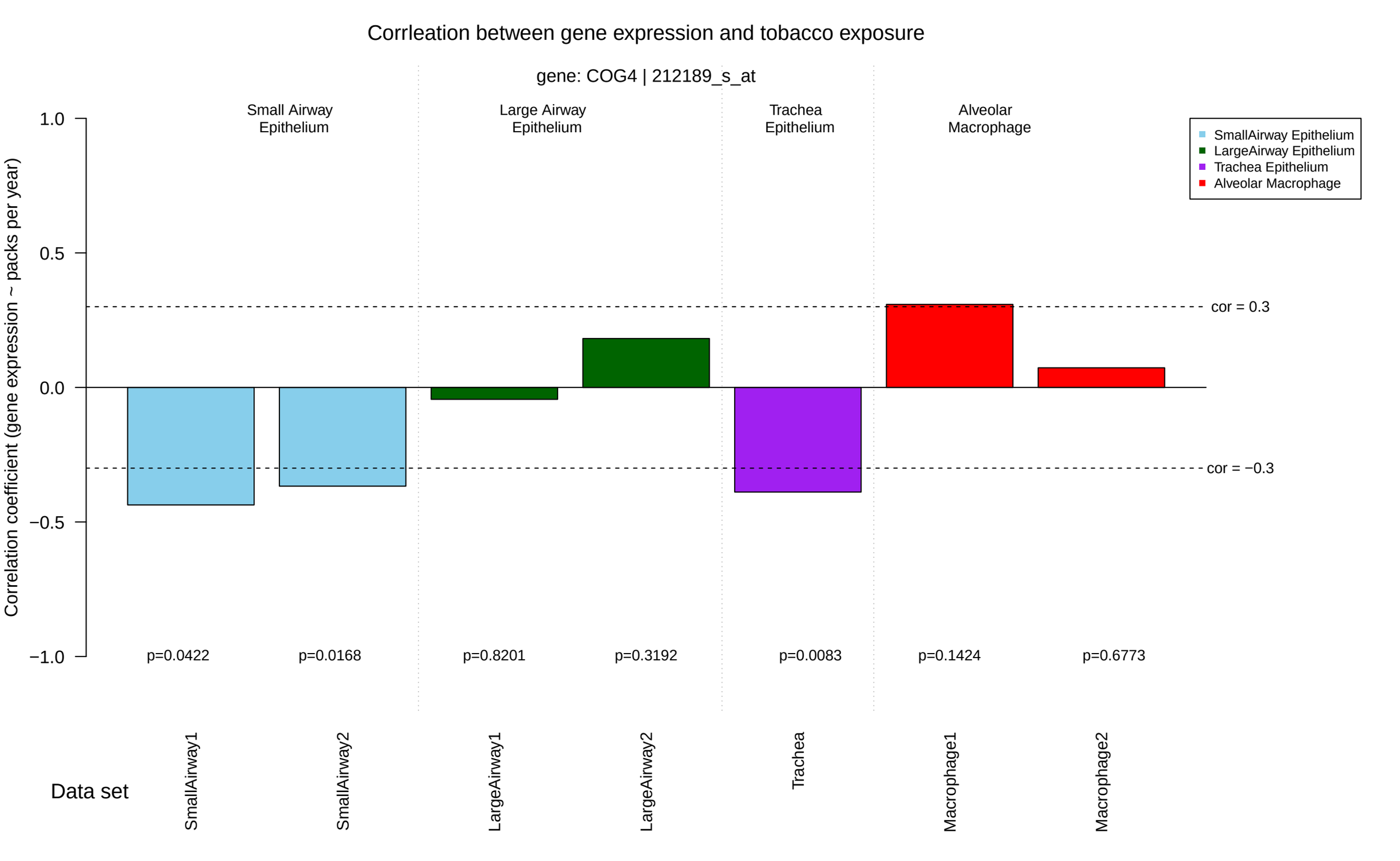

set: LargeAirway1 (p=0.2249)
COG4 | 212189_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5071)
COG4 | 212189_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.008)
COG4 | 212189_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.359)
COG4 | 212189_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.622)
COG4 | 212189_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COG4 | 212189_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0422
p=0.0168
p=0.8201
p=0.3192
p=0.0083
p=0.1424
p=0.6773
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2